



Research article

Feasibility assessment of Japan's fertilizer reduction target: A meta-analysis and its implications for nitrogen waste

Kazuya Nishina^{a,b,*}, Kentaro Hayashi^b, Azusa Oita^{c,b}, Kei Asada^c, Atsushi Hayakawa^d, Tomohiro Okadera^a, Takashi Onodera^{a,b}, Tatsuya Hanaoka^a, Kazuaki Tsuchiya^a, Koya Kobayashi^e, Nobuhisa Koga^f

^a National Institute for Environmental Studies, 16-2, Onogawa, Tsukuba, 305-8506, Ibaraki, Japan

^b Research Institute for Humanity and Nature, 457-4, Kamigamo-Motoyama, Kita-ku, Kyoto, 603-8047, Kyoto, Japan

^c Institute for Agro-Environmental Sciences, National Agriculture and Food Research Organization, 3-1-3, Kannondai, Tsukuba, 305-8517, Ibaraki, Japan

^d Akita Prefectural University, 241-438, Kaidobashi-Nishi, Shimoshinjo-Nakano, Akita, 010-0195, Akita, Japan

^e University of Tsukuba, 1-1-1, Tennodai, Tsukuba, 305-8577, Ibaraki, Japan

^f Kyushu Okinawa Agricultural Research Center, National Agriculture and Food Research Organization, 2421, Suway, Koshi, 861-1192, Kumamoto, Japan

ARTICLE INFO

Dataset link: [10.6084/m9.figshare.26808307](https://doi.org/10.6084/m9.figshare.26808307), <https://figshare.com/s/97b9008d63eb5839cd57>

Keywords:

Nitrogen waste
Synthetic nitrogen fertilizer
Reduced fertilizer management
Meta-analysis
Yield response
Linear hinge model

ABSTRACT

Reducing synthetic nitrogen fertilizer (SNF) use is critical for mitigating environmental nitrogen (N) pollution caused by excessive reactive nitrogen and lowering the cost of chemical N fertilizer inputs in agricultural management. In response, Japan has set a national target to reduce chemical fertilizer inputs, with a goal of 30% reduction in SNF use. To assess the feasibility of this target, we collected ca. 1700 records from domestic SNF reduction experiments and conducted a meta-analysis to examine the effects of SNF reduction on crop yield. Using a linear plateau response model of yield to SNF application in the experiments, we assumed that the conventional SNF rates used in control plots were sufficient to reach maximum yield. Our meta-analysis indicates that a 30% SNF reduction in SNF is achievable without significant yield loss on average. This reduction can be facilitated through the use of functional fertilizers and replacement of SNF with organic materials. Additionally, we evaluated changes in nitrogen waste from domestic cropland under the new national target using simple scenarios. Our results show that a 30% SNF reduction, without substituting by organic materials, would only decrease N waste by 8.8%. However, our meta-analysis finds that greater reductions in nitrogen input are possible, indicating potential for further optimization of N management, including the use of organic amendments.

1. Introduction

The global biogeochemical nitrogen cycle is currently exceeding its critical limits, primarily due to excessive anthropogenic nitrogen loading, which is approaching planetary boundaries (Richardson et al., 2023). The critical threshold for nitrogen flows is estimated at 62 Tg-N/year, primarily concerning coastal eutrophication. However, as of 2020, approximately 100 Tg-N of nitrogen is being deposited on land, mainly through the application of chemical fertilizers synthesized via the Haber-Bosch process (Tian et al., 2022a). Additionally, another 65 Tg-N per year of reactive nitrogen is being deposited onto terrestrial ecosystems through atmospheric deposition (Tian et al., 2022a). Excess reactive nitrogen has detrimental environmental effects on air, water, soil, and climate, necessitating urgent measures to mitigate these impacts (Sutton et al., 2013). In response to these concerns, the issue

of reactive nitrogen has gained recognition as a global environmental problem. This led to the adoption of a resolution on sustainable nitrogen management (UNEA/EA.4/Res.14) by the Fourth United Nations Environment Assembly, identifying the need to reduce emissions of anthropogenic nitrogen on a global level. Along with the establishment of an international framework, efforts to reduce 'nitrogen waste' in each country are expected to accelerate until 2030 (Sutton et al., 2021). Nitrogen waste is defined as the total of artificial nitrogen losses to the atmosphere and water (including N₂ losses), and agriculture plays a major role in nitrogen waste (Leip et al., 2022). Additionally, the Kunming-Montreal Global Biodiversity Framework, adopted under the Convention on Biological Diversity in 2022, includes Target 7, which aims to reduce excess nutrient loading to protect biodiversity (The Convention on Biological Diversity, 2022). Recent research indicates that the agricultural sector could reduce nitrogen fertilizer usage by

* Corresponding author at: National Institute for Environmental Studies, 16-2, Onogawa, Tsukuba, 305-8506, Ibaraki, Japan.
E-mail address: nishina.kazuya@nies.go.jp (K. Nishina).

<https://doi.org/10.1016/j.jenvman.2024.123362>

Received 8 September 2024; Received in revised form 28 October 2024; Accepted 12 November 2024

Available online 30 November 2024

0301-4797/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

20 Tg-N globally, potentially lowering the environmental impact of nitrogen fertilizers by more than 30% (Gu et al., 2023). We need to further accelerate our efforts to reduce waste nitrogen on a global scale.

In Japan, crop production is a significant source of reactive nitrogen emissions. According to Hayashi et al. (2021), the current use of synthetic N fertilizer (SNF) in Japan was about 300 Gg-N per year, which contributes much of the reactive nitrogen pollution in Japan. In 2022, the Japanese Ministry of Agriculture, Forestry and Fisheries (MAFF) introduced the Act to Promote Low Environmental Impact Business Activities for the Establishment of Environmentally Harmonized Food Systems, commonly known as the MIDORI Act. This initiative, similar to 'Farm to Fork' strategy under the European Green Deal, targets a reduction of domestic chemical fertilizer use by 20% by 2030 and 30% by 2050 while promoting organic farming across approximately 25% of all farm land in Japan (OECD, 2023). However, to the best of the author's knowledge, this ambitious goal has not yet been scientifically evaluated in terms of its impact on crop productivity, leaving open questions about its feasibility.

The recent surge in SNF prices has provided additional motivation for Japanese farmers to reduce their reliance on chemical nitrogen fertilizers (Snapp et al., 2023). According to the Japan Agricultural Cooperatives (JA, 2024), urea prices were temporarily 2.5 times higher in 2022 than before 2020, and they are expected to remain more than 1.5 times higher in 2023. In response to these economic pressures, MAFF has implemented a subsidy program to support farmers' efforts to reduce fertilizer use by 20%. Despite these incentives, the nitrogen use efficiency (NUE) of Japanese cropland remains low compared to the global average (Zhang et al., 2021; Lassaletta et al., 2014), and the amount of fertilizer applied per unit area in Japan is higher than in many other countries (Tian et al., 2022a). Consequently, there is a pressing need to reduce chemical fertilizer use not only for environmental reasons but also due to economic constraints. Furthermore, achieving nitrogen fertilizer savings without reducing yield leads to an improved level of fertilizer use from an economic perspective.

However, implementing a national reduction target poses significant challenges at the farm level, given the wide variety of agricultural products produced across Japan. The feasibility of implementing such a target on a national scale has not been thoroughly verified. Nonetheless, numerous nitrogen fertilizer reduction trials have already been conducted across Japan, in academia and at national and regional agricultural experiment stations. We hypothesize that an integrated interpretation of these experiments—specifically through meta-analysis—would be valuable in assessing the average reduction rates that could be achieved in line with national goals. Several yield response functions can be assumed when examining the relationship between fertilizer inputs and crop yields, with many major models predicting a plateau at high fertilizer application rates (Dhakal and Lange, 2021). This plateau is based on Liebig's law of the minimum, where other factors become limiting once nitrogen application rates are sufficiently high (Dhakal and Lange, 2021).

In this study, we estimated the rate of SNF reduction that would not affect crop yield by examining the relationship between the reduction rate and yield in domestic nitrogen fertilizer reduction experiments. To achieve this, we applied a hinge linear model in our meta-analysis, which corresponds to the linear-plateau model commonly used in crop modeling (Elder and Fong, 2019; Dhakal and Lange, 2021). Our goal was to evaluate the feasibility of new national reduction targets for synthetic chemical fertilizers in Japan by determining the average SNF reduction rate that would maintain maximum yield across a pooled dataset. Additionally, we assessed the impact of the MIDORI Act's targets on national nitrogen waste in Japanese cropland. Based on these analyses, we discussed the challenges of transitioning to more sustainable agricultural practices in Japan, particularly concerning N waste.

2. Material and methods

2.1. Literature survey

We conducted a systematic literature survey using Google Scholar and AgriKnowledge (<https://agriknowledge.affrc.go.jp>) to identify studies related to SNF reduction experiments in Japan. AgriKnowledge contains reports from agricultural experiment institutes owned by local governments in Japan, allowing us to gather both peer-reviewed scientific papers and local institute reports. The literature was collected in both English and Japanese using search terms including, but not limited to, "synthetic nitrogen fertilizers", "Japan", "crop yield", and "reduction". Boolean operators (AND, OR) were used to refine the search queries.

The information extracted from the literature includes author names, publication years, journal names or publishing institutions, cultivated crops, cultivation periods, SNF input in control areas, SNF input or organic material input in reduced fertilization areas, and organic material N efficiency. Unfortunately, many reports from local institutes did not include standard deviations in their results, even though treatments were replicated.

We categorized nitrogen fertilizer reduction methods into four approaches. The first is the simple nitrogen reduction approach, referred to as "Reduction". This category involves nitrogen fertilizer reduction without any change in fertilizer type, e.g., optimized by considering the timing and placement of application. This category also includes the so-called zero N treatment, where no chemical fertilizers were applied at all. The second category, "Functional fertilizer", involves changing from ordinary chemical fertilizers to functional fertilizers, such as slow-release fertilizers, chemical fertilizers with nitrification inhibitors, or lime nitrogen. The database included only a few records for nitrification inhibitor fertilizers, with most records pertaining to coated fertilizers. The third category, "Organic materials", includes the partial or full replacement of SNF with organic materials such as animal manure, meat and bone meal, crop residues, and other organic amendments. The fourth category is "Green manure", which includes legumes (e.g., hairy vetch, crotalaria), rye, and Italian ryegrass, which are plowed into the soil after being grown. Nitrogen input in the organic material treatments was calculated based on the N-efficacy of organic materials during cultivation. We then calculated the ratio of fertilizer in the control and organic amendment treatments as the total N input. For those references that reported N efficacy of organic materials in the literature, we used those values directly. For references that only reported the nitrogen content of the organic material, we used a fertilizer efficacy rate of 0.3, as recommended in previous literature (Nagano prefecture, 2023). Finally, for references that only reported the amount of organic material inputs, we used a web application that calculates the N efficacy of organic materials (Koga et al., 2023a). This web app estimates the amount of N mineralization during the cultivation period based on weather information at any location in Japan, using the growing schedule, the amount of organic materials, and organic material type as input data (Koga et al., 2023b). The app considers 10 types of organic material, including cattle manure, swine manure, chicken manure, vegetable oil meal, fish meal, bone meal, rice bran, green manure, crop residue, and food compost (Koga et al., 2023a). These estimates of nitrogen efficacy for organic materials were also made available in the database.

2.2. Data analysis

A meta-analysis was conducted to synthesize the findings from the included studies. The analysis was performed using the statistical software R. We used odds ratios of yields in each crop product as effect sizes of the reduction treatments to quantify the impact of SNF.

We applied the hinge linear model (Elder and Fong, 2019) to infer breakpoints for the yield response to the percentage of SNF reduction,

corresponding to the linear-plateau model used in crop models (Dhakal and Lange, 2021). Generally, the SNF application rate in the control plots of domestic fertilizer reduction experiments is assumed to be the amount of fertilizer typically applied in the region. In some cases, it corresponds to the recommended fertilizer application rate issued by local authorities. Therefore, we considered the control plots in fertilizer reduction studies as representative of current SNF use in Japan. The yields in the control plots were assumed to be maximized, and we did not expect any issues related to excessive fertilizer application. The amount of fertilizer applied in the control plots was presumed to be higher than necessary. Indeed, the nitrogen use efficiency (NUE) of Japanese cropland is relatively low (Hayashi et al., 2021) compared to the global average (Zhang et al., 2021; Lassaletta et al., 2014), while the amount of fertilizer applied per unit area is higher than in other countries (Tian et al., 2022a). The hinge linear model is defined as follows;

$$Y_{ij} = \beta_0 + (\beta_0^* + \beta_1(X_{ij} - X^*))D_{ij} + e_{ij} \quad (1)$$

Y_{ij} indicates log yield response ratio, which is defined as:

$$Y_{ij} = \ln \left(\frac{\text{Yield}_{T_{ij}}}{\text{Yield}_{C_{ij}}} \right) \quad (2)$$

$\text{Yield}_{T_{ij}}$ and $\text{Yield}_{C_{ij}}$ denote the yield in each treatment and control plot, respectively. β in Eq. (1) are the coefficients of the linear model. X_{ij} indicates the percentage of SNF reduction in the treatment plot relative to the control plot in each study, with i and j representing the treatment ID and literature ID, respectively. The ‘literature ID’ is a number assigned to each report in the database to distinguish between literatures. The ‘treatment ID’ is automatically assigned to each treatment (excluding the control plot) in each paper (j). e_{ij} is an error term. D_{ij} is defined as:

$$D_{ij} = \begin{cases} 1 & X_{ij} \geq X^* \\ 0 & X_{ij} < X^* \end{cases} \quad (3)$$

where X^* is the breakpoint for the hinge regression line, and this parameter is estimated from the data. When fitting the model, we considered the weight of each reported value. However, most studies did not report variance measures (or sample sizes) for yield, particularly in institutional reports. Following the approach of previous literature (Chen et al., 2018), we used sample size as the weight in the regression instead of variance since variance measures were often missing. The sample size weight function used was according to Adams et al. (1997):

$$w_{ij} = \frac{N_{ij}^C N_{ij}^E}{N_{ij}^C + N_{ij}^E} \quad (4)$$

where N_{ij}^C and N_{ij}^E are the numbers of replications in the control and treatment plots, respectively. To estimate these parameters by the maximum likelihood estimation, we used the R package ‘‘chnp’’ (Fong et al., 2017). For organic materials, we performed the same hinge regression analysis after switching X_{ij} in Eq. (1) to the relative nitrogen input in the control plot, which is the sum of the SNF and organic N amendment based on N efficacy (or the amount of N derived from 100% organic materials) as described in Section 2.1.

2.3. Scenario analysis on N waste in cropland

To assess the amount of N waste in a Japanese cropland, we used the Coupled Human and Natural System (CHANS) model originally developed by Gu et al. (2013). The CHANS model integrates bottom-up N input–output fluxes and top-down measurements of reactive nitrogen (Nr) fluxes across 14 subsystems as a pool. These subsystems consist of cropland, livestock, industry, humans, pets, waste and wastewater treatment, and the environmental components (surface water, atmosphere). At the national scale, each annual flow is estimated using activity data such as national statistics and emission factors. The

Table 1

Number of records collected for the meta-analysis by reduction method and crop type.

Crop/Treatment	Reduction	Organic	Slow-release fert	Green manure	Total
Cereals*	111	109	55	16	295
Rice	137	218	141	65	561
Green vegetable	133	165	100	55	453
Root vegetable	105	69	4	0	178
Other vegetable	67	21	21	12	122
Orchard	22	19	4	0	45
Tea	18	0	21	0	39
Pasture	16	22	5	0	43
Total	613	624	351	148	1736

CHANS-Japan model is a revised version of the CHANS model adapted for use in Japan (Hayashi et al., 2021).

Using the CHANS-Japan model from Hayashi et al. (2021), we evaluated N flows in Japanese cropland under two scenarios, comparing them to current N flows (from 2015) to explore how cropland-derived waste might change as a result of the Midori Target. In the first scenario, ‘SNF to Org’, a 30% reduction in SNF is offset by the application of organic materials on an equivalent nitrogen basis, keeping the total N input to cropland the same as under current conditions. This is based on the assumption that the ‘‘organic materials’’ category of methods for reducing chemical fertilizer use is used. However, the assumptions regarding the effectiveness of the fertilizer are optimistic. This is based on the assumption that there will be an effect from the repeated use of organic materials. The second scenario, ‘SNF reduction’, involves a 30% net reduction in N inputs achieved through optimization of fertilizer use (in terms of rate, placement, and timing) or by utilizing functional fertilizers such as slow-release fertilizers. This is based on the category of reduction methods in this study, which are ‘‘Reduction’’ or ‘‘Functional Fertilizer’’ in meta-analysis. Regarding green manure, there was a lack of data on nitrogen input in the meta-analysis, and it was difficult to quantitatively estimate nitrogen input, so this scenario did not explicitly take it into account. To evaluate N gas emissions, we updated the emission factors for nitrous oxide (N_2O) (Akiyama et al., 2023), nitric oxide (NO) (Nishina et al., 2012), and gaseous nitrous acid (HONO) (Xue et al., 2022) from those used in Hayashi et al. (2021).

3. Results and discussion

3.1. Reduction potential and national target

From the domestic field experiment literature (252 reports; 72 are peer-reviewed papers and the left consists mainly of reports from the Agricultural Experiment Station), we collected 1736 records for fertilizer N-reduction treatments (Table 1). The geographical distribution of records for reduction treatments generally corresponds to agricultural activity in Japan (Fig. 1). Hokkaido (Japan’s northernmost prefecture) has the largest records in with a record number of 211 in our database. For example, ammonium sulfate, the most commonly used nitrogenous fertilizer in Japan, accounted for 41.1% of the total shipment value in Hokkaido alone in 2015 (Association of Agriculture and Forestry Statistics, 2023). Unfortunately, there are three prefectures for which we have no records at all (Fig. 1), but we have collected the records for neighboring prefectures with similar climates and agricultural products, and the database created in this study might generally reflects the agricultural managements in Japan without strong biases. As a whole database, the SNF reduction rates range from 6.6% to 100%, with a median and mean of 50% and 55.6%, respectively. The periods for the reduced N fertilizer trials in the database range from 1980 to 2020, with a median of 2006, and approximately 80% of the records are from 2000 onwards. Since the database covers a wide range of cultivars, there is a large variation in the amount of fertilizer used for each variety. On the other hand, the sample size for paddy rice was large (Table 1) because rice is a major crop in Japan. So, we focused on paddy rice for

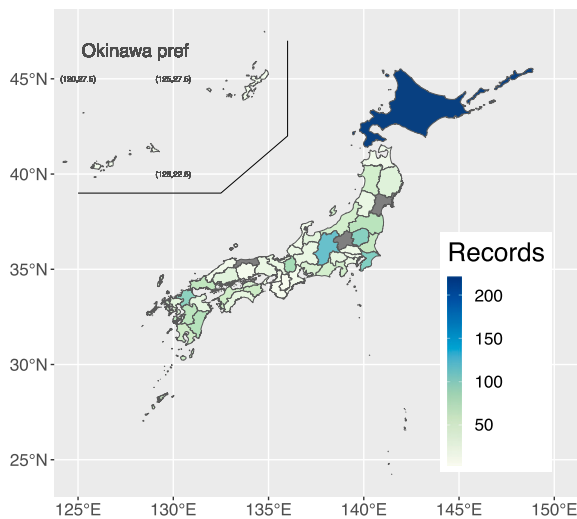


Fig. 1. Map showing the distribution of database records for SNF reduction treatments across prefectures Japan.

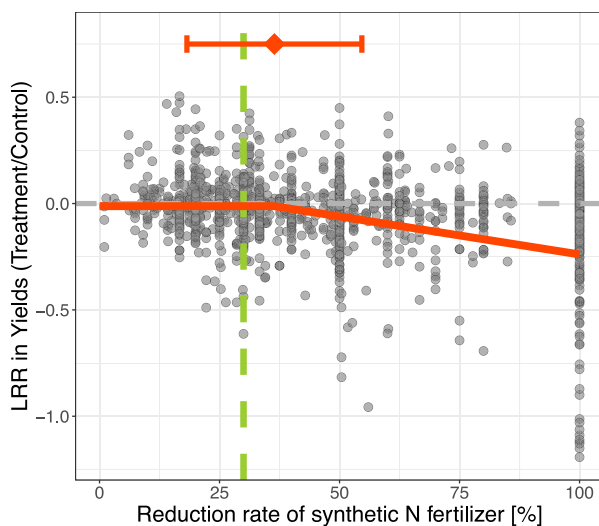


Fig. 2. Relationship between the log yield ratio and the rate of synthetic nitrogen fertilizer (SNF) reduction. Each dot represents a record, with the red line indicating the fitted hinge regression. The orange square and error bar denote the estimated breakpoint in the hinge regression model and its standard error. The green dashed line marks the Midori Target of a 30% reduction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

N application rate. For rice paddy, the mean ($\pm$ SD) of nitrogen input in the control plot (referred to as the conventional plot in many studies) used in this study was 85.9 ($\pm$ 24.0) kg-N/ha per growing season ($N = 561$). The fertilizer rate recommended by MAFF for rice paddy varies with rice variety and growing environment, but it is between 60 and 100 kg-N/ha. In addition, the average nitrogen input for rice cultivation in Japan, as based on national statistics, was estimated to be 66.3 kg-N/ha for chemical fertilizer in 2005 (Mishima and Kohyama, 2010) and 25 kg-N/ha for organic materials in 2005 (Mishima et al., 2012), respectively. The sum of these two sources was approximately the same as the average value in this study.

The estimated breakpoint for the segmented regression in the overall pooled analysis was 37.3% (Fig. 2). The fitted hinge regression model showed a significantly negative trend (β_1 was -1.78×10^{-3}) beyond the breakpoint. This result suggests that yield response is less sensitive until a 37.3% SNF reduction is achieved. As described in the introduction, we hypothesized that this non-sensitive range corresponds

to the plateau in the linear-plateau response model, a major nutrient-crop yield response model (Dhakal and Lange, 2021). Therefore, it is presumed that reducing fertilizer use by 30% can be achieved without significant yield loss on average in Japan. Regarding the national ‘Midori Target’ of a 30% reduction in chemical fertilizers from the perspective of SNF reduction, our results support its feasibility with the currently available agricultural management options.

In the next step, we split the dataset by each reduction method and fitted the hinge regression model for each method (Fig. 3). For the ‘Reduction’ method, the estimated breakpoint was 46%, which is still higher than the reduction target (30%) in Midori (hereafter termed “the Midori target”), however, the linear slope after the breakpoint was steeper than in the pooled analysis (Fig. 2). For the ‘Functional fertilizer’ method, the estimated breakpoint (16.7%) was lower than the Midori target (Fig. 3). However, the regression slope beyond the breakpoint was relatively small, though significantly negative ($p < 0.05$), indicating that the effect of SNF reduction on yield was less sensitive even beyond the Midori target. For ‘Organic materials’ and ‘Green manure,’ the breakpoints were 84.8% and 80.0%, respectively. Note that for organic material and green manure amendments, the reduction in chemical fertilizers does not correspond to a reduction in nitrogen input, as there is an independent input of nitrogen derived from organic materials. Therefore, a further analysis of the relationship between compost substitution and yield was conducted using total nitrogen input as an explanatory variable (Fig. 4). The data were further divided to account for the number of years of organic material application in the fertilizer reduction studies, as the records in the database included long-term continuous application experiments aimed at improving soil fertility. The estimated breakpoints for the relative total nitrogen input were 60.0%, 56.6%, and 22.5% for single-year (or unknown), multi-year, and long-term continuous application, respectively (Fig. 4), indicating that a net reduction in nitrogen input is possible through the combined use of chemical fertilizers and organic materials or the application of organic materials alone. Although the yields in long-term organic amendments did not appear to decrease with a lower nitrogen input rate (Fig. 5), this study considers that simple conclusions should be avoided due to the small number of samples, especially for the long-term continuous use treatments.

The longer the continuous use of organic materials, the lower the nitrogen input required to achieve stable yields (Fig. 4). However, this study cautions against drawing simple conclusions due to the small sample size, particularly regarding long-term continuous use ($N = 87$). It is noteworthy, though, that this yield-stabilizing trend aligns with results from global meta-analyses of long-term continuous use trials (Chen et al., 2018). In the multi-year category, there are records of yields being maintained despite lower nitrogen inputs. This may be because nitrogen fertilization in this study only considered the crop growth period in the current year (Koga et al., 2023b), and the fertilization estimates themselves, based on the nitrogen efficacy rate or the model in the app (Koga et al., 2023b), may be underestimates in some experiments.

Our results showed a lower sensitivity of crop yields in Japanese croplands to reductions in nitrogen fertilizer application, suggesting that current nitrogen fertilizer rates in Japan are generally sufficient. Our meta-analysis indicates that a 30% reduction in SNF, as targeted by the MIDORI Act, is fully feasible even with simple optimization of fertilizer application. This may be due to the high soil fertility in Japanese farmland (at least in the study areas) and the fact that more than enough nitrogen fertilizer is typically applied in conventional management. In fact, the average NUE (the ratio of produced crop N to total input N (= fertilizer N + manure N + BNF + N deposition + irrigation N)) for overall domestic crop production in Japan is 29.5% (Hayashi et al., 2021), which is lower than the global average of 47% (Lassaletta et al., 2014; Zhang et al., 2021), indicating substantial room for improvement in NUE in Japan.

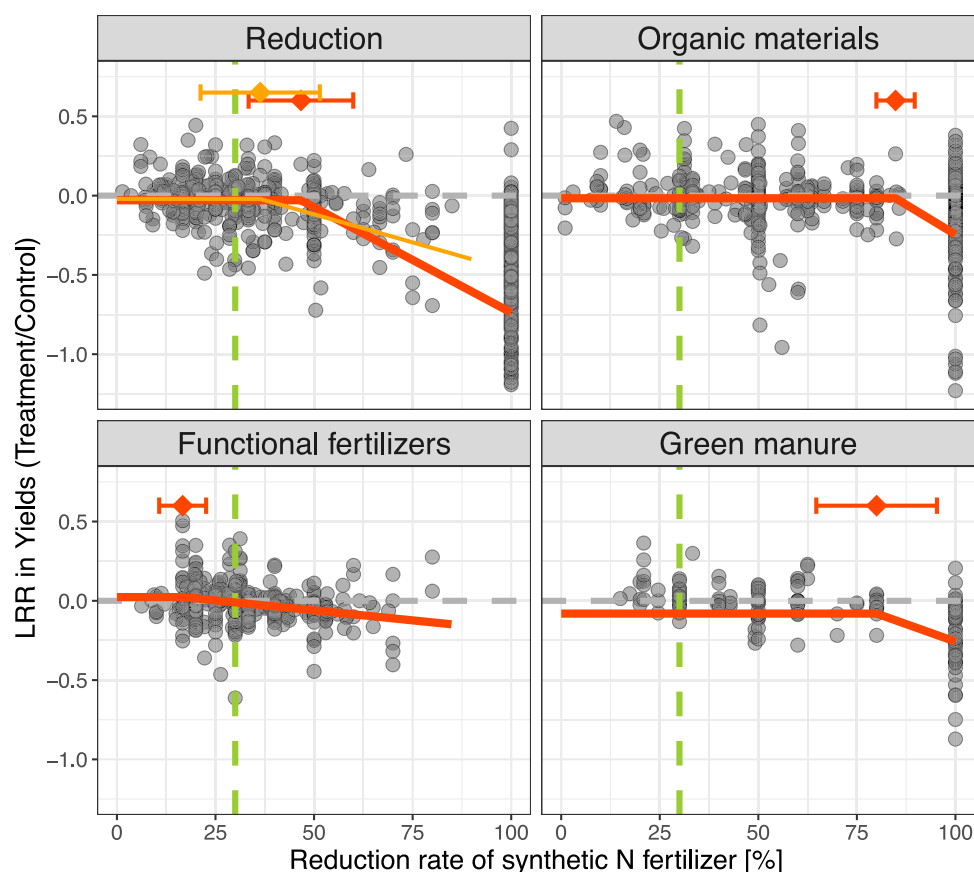


Fig. 3. Relationship between the log yield ratio and the rate of synthetic nitrogen fertilizer (SNF) reduction by method. The orange lines indicate the fitted hinge regression, excluding records from zero-N treatments. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

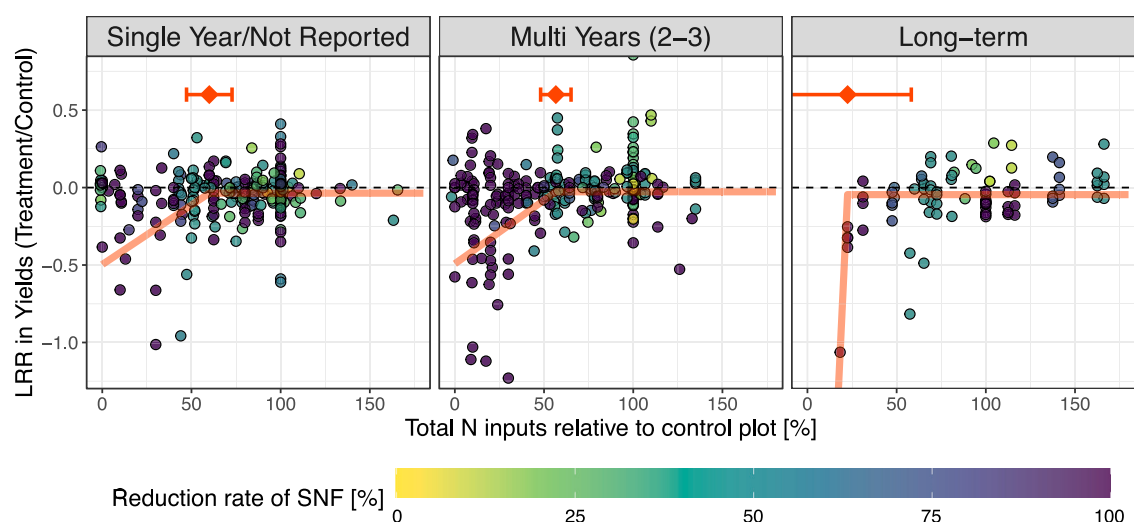


Fig. 4. Relationship between relative total nitrogen input and log relative yield ratio in organic fertilizer amendments. The colors of the dots indicate the rate of SNF reduction relative to the control plots, which correspond to the X axis in Fig. 3. The total N input is the sum of SNF and organic material amendments, or organic material amendments only in the case of 100% SNF reduction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

For example, in carrot cultivation, efforts have been made to maintain yields and minimize groundwater impacts by assessing the amount of residual inorganic nitrogen in the soil before fertilizer application and adjusting nitrogen rates accordingly (Yamaki, 2011). Due to the limited records of N content in the harvested fraction in our database, we did not conduct further analyses for NUE in this study. Instead, we focused on yield ratios in the database for SNF reductions of 30% or more. The results showed that 25.5% of the respondents maintained

or increased yields with simple fertilizer reductions, while slow-release fertilizers had the highest success rate at 40.2%, followed by organic materials at 36.7% (35.1% for green manure). These management practices could therefore enhance NUE and yield. Notably, slow-release fertilizers tended to show relatively lower variation in yield response within this reduction range compared to other categories, likely due to their ability to provide consistent N fertilizer efficacy regardless of climate conditions (Beig et al., 2020).

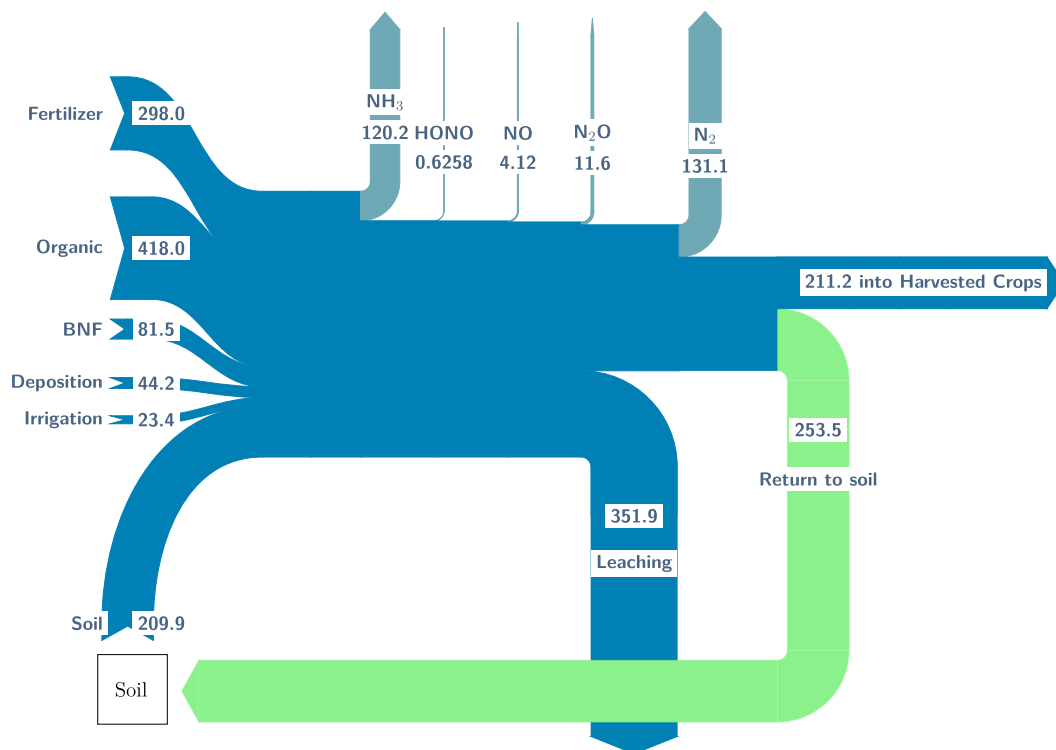


Fig. 5. Diagram of N flows in Japanese cropland. The widths of the arrows are proportional to the magnitude of annual N flows (Gg-N Year⁻¹) at 2015. Fertilizer, Organic, BNF, and Deposition indicate SNF, organic materials, biological N fixation, and atmospheric N deposition, respectively. Irrigation indicates N input from irrigated water bodies.

As already mentioned, policies with specific numerical targets for reducing chemical fertilizers, such as Europe's Farm to Fork Strategy, are beginning to emerge. Such a trend is expected to accelerate in the future in response to the UN resolutions (UNEA/EA.4/Res.14 and UNEP/EA.5/Res.2). We believe that a meta-analysis approach in this study, which summarizes existing fertilizer reduction experiments, can provide a scientific assessment of the feasibility of such reduction targets in countries and regions where excessive fertilizer application is practiced (Hijbeek et al., 2017; Jiang et al., 2018; Sutton et al., 2017; Tian et al., 2022b). In particular, optimization of chemical fertilizer inputs is one of the most efficient and side-effect-free counter measures to reduce nitrogen waste in agricultural land (Brownlie et al., 2024), and it would be useful to find a reduction rate that can maintain yields on average at national/regional scales. The same approach could be applied not only to nitrogen, but also to phosphorus and potassium fertilizers.

3.2. Consequences of the SNF reduction on domestic N waste

The current annual cropland N flows are displayed as a Sankey diagram in Fig. 5, which is based on Hayashi et al. (2021) and updated emission factors for some fluxes. The N input by SNF was 298 Gg-N year⁻¹, whereas the organic N input was 418 Gg-N year⁻¹. Thus, the N input from organic materials exceeded that from chemical fertilizers. From 2000 to 2006, the two were almost equal in amount (Shindo et al., 2009; Hayashi et al., 2021), but after 2007, the chemical fertilizer contribution gradually decreased, and the use of organic material became relatively higher on an N basis (Hayashi et al., 2021). However, it is important to note that organic amendments (mainly animal manures) in Japanese farmers are generally applied to the soil for the purpose of increasing soil fertility, i.e. to improve the physical properties of the soil, such as its water retention capacity, and its ability to retain nutrients and not primarily for nitrogen fertilization (especially with cattle manure). This suggests that there is significant room in Japan to utilize organic materials to meet the nitrogen requirements of crops and

improve Japan's low NUE. In fact, the nitrogen fertilizer dependency (FDN: the ratio of chemical fertilizer to anthropogenic nitrogen input to agricultural land) proposed by Quemada and Lassaletta (2024) was 41.6% in Japan in 2015, which is lower than the world average (82.4%) (Quemada and Lassaletta, 2024).

Under the 'SNF to Org' scenario, ammonia (NH₃), N₂O, and N₂ emissions are higher than in the current scenario because organic materials have higher emission factors than those in SNF (Fig. 6). For NH₃ volatilization, the major emitted gas from cropland (Fig. 5), annual emissions increased from 134.4 Gg-N year⁻¹ in the current scenario to 155 Gg-N year⁻¹ in the 'SNF to Org' scenario. This indicates that a simple replacement of SNF with organic nitrogen may increase some nitrogen emissions compared to the 'SNF reduction' scenario, where nitrogen emissions to air and water are reduced in proportion to the net reduction in nitrogen inputs (560 Gg-N year⁻¹ in the current scenario and 511 Gg-N year⁻¹ in the 'SNF reduction' scenario). It should be noted that optimal organic fertilizer management will not necessarily lead to greater environmental benefits (Tuomisto et al., 2012). Therefore, when organic materials are used intensively, it may be necessary to introduce management practices that mitigate the release of additional reactive Nr into the environment. For example, deep fertilization with animal manures can be effective in reducing NH₃ volatilization (Liu et al., 2023).

Halving N leaching through SNF reduction alone, as targeted by the Midori Act, is challenging. This difficulty is similarly noted in the EU's Farm to Fork Strategy (Billen et al., 2024). However, if more ambitious reductions are implemented beyond the Midori target, the future nitrogen balance in cropland could improve significantly, potentially falling below the levels projected under the SNF reduction scenario. As previously mentioned, in Japan, the primary goal of using livestock manure and other organic materials has been to enhance soil quality, rather than optimizing nitrogen fertilizer effectiveness. Our meta-analysis indicates that a net reduction in nitrogen inputs is achievable through a combination of chemical fertilizers and compost (Fig. 4). This suggests that optimizing both newly adopted organic farming practices under

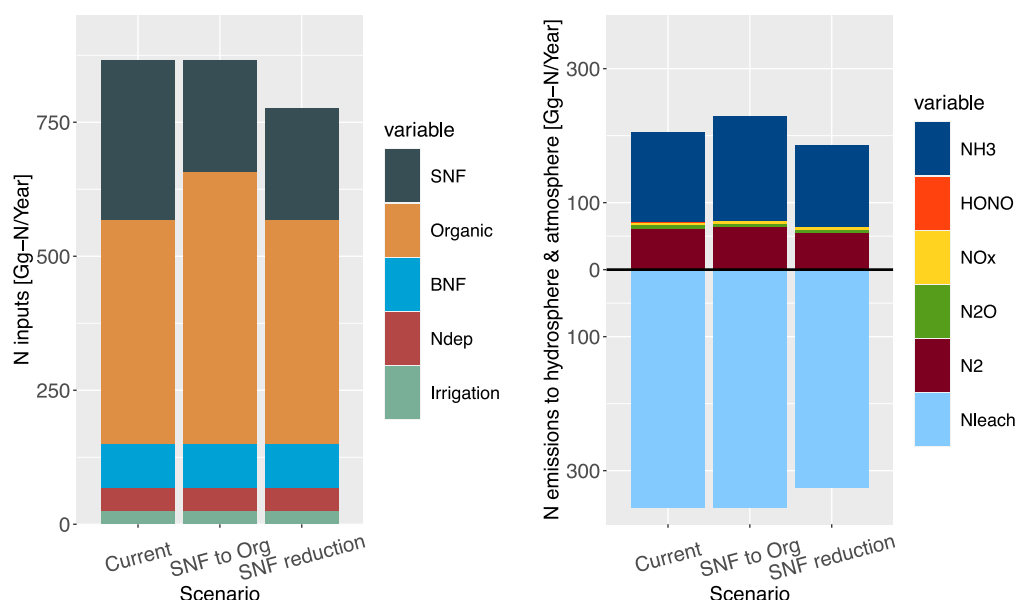


Fig. 6. Comparison of N inputs (left) and N waste (right) in Japanese cropland under current conditions and in simple scenarios. The current N inputs and N waste correspond to the data in Fig. 5.

the Midori target and existing applications of organic materials, such as compost, based on their nitrogen efficacy, can lead to reduced nitrogen inputs and lower waste nitrogen emissions. In Japan, the development of Internet of Things (IoT) technology, including smartphone apps, for farmers has provided tools for predicting the nitrogen fertilizer efficacy of organic materials such as livestock manure, vegetable oil cake, and coated urea fertilizer, according to soil conditions like temperature. This technology could significantly enhance nitrogen management in soils, leading to more efficient and sustainable agricultural practices.

3.3. Barriers for N pollution abatement in Japanese cropland

The nitrogen fertilizer reduction methods discussed in this study fall within Tiers 1 and 2 as defined by Gu et al. (2023), representing primarily traditional practices that are relatively easy for farmers to adopt independently. However, large-scale changes to farmland, such as infrastructure development or the establishment of vegetable factories, were not included in our analysis. Japanese cropland management predominantly consists of smallholder farmers, similar to the situation in China (Ren et al., 2019). Although the government promotes the establishment of village farming corporations, which could lead to the consolidation of farmland and production rationalization (Fuhmann-Aoyagi et al., 2024), such large-scale transformations were outside the scope of this study. Additionally, certain newly developed techniques for saving SNF were not covered in our analysis. For instance, precision fertilization technology, which incorporates Global Navigation Satellite System (GNSS) and drip irrigation fertilization (only one study of which is included in our database), may contribute to reducing nitrogen fertilizer application. Further optimization could be achieved by combining these technologies with existing nitrogen reduction strategies. However, more localized analyses, similar to those conducted in the EU (Sanz-Cobena et al., 2023), are essential to understand the specific needs and challenges in different regions of Japan.

Organic farming is also a significant focus of the Midori strategy (OECD, 2023). As in the EU (Billen et al., 2024), Japanese livestock production relies heavily on imported feed. The livestock industry is geographically dispersed in Japan (Leon et al., 2012), so it is likely that some prefectures will be able to offset 30% of chemical fertilizers reduction with organic matter on N basis supplied within the prefecture, while others will not. Farm management-level challenges also persist, such as the dependency of crop productivity on soil fertility factors

including phosphorus, potassium, and other nutrients, in addition to nitrogen (Chen et al., 2018; Wu et al., 2020). While sewage sludge compost is recommended within the context of a circular economy, the database does not include data from experiments on sewage sludge compost. Unfortunately, the lack of detailed information in the collected literature prevented a deeper analysis of these aspects in this study.

Slow-release fertilizers offer a promising option for reducing nitrogen inputs in Japanese agriculture, especially since Japan is located in monsoon Asia and tends to experience high rainfall during the growing season (Shoji et al., 2001). However, functional fertilizers, including slow-release types, are typically more expensive than standard SNFs, such as urea (Beig et al., 2020). For example, coated urea costs approximately three times as much as regular urea in Japan. Slow-release fertilizers can reduce labor costs in agricultural management, which may offset the higher costs of functional fertilizers in certain crops. However, the economic feasibility of these fertilizers highly depends on the type of crop and specific agro-environmental conditions (e.g., Valkama et al., 2013). Currently, the share of slow-release nitrogen fertilizers in domestic SNF production and Japan's SNF imports is only about 9% (Association of Agriculture and Forestry Statistics, 2023), indicating potential for increased use in the future. Therefore, considering the profitability of each crop is essential. Another critical aspect is that the use of plastic films for coating urea contributes to plastic pollution in watersheds and coastal areas (Katsumi et al., 2020). Most of the slow-release fertilizers analyzed in this study were of the sigmoid release type, which uses plastic materials. It is important to note that further research is needed on alternative types of slow-release fertilizers. Climate change is expected to increase the frequency of heavy rainfall during crop rotation periods in Japan (Imada et al., 2020; Mori et al., 2021), which could negatively impact NUE in crops (Lassaletta et al., 2023). To prepare for future climate changes, adaptive nitrogen management practices should be developed to address shifts in temperature and precipitation patterns. However, there are still limited domestic studies investigating the effects of climate change and elevated CO₂ levels on nitrogen management in crop production (Tang et al., 2021; Hayashi et al., 2022).

3.4. Conclusions

Our meta-analysis of fertilizer reduction experiments suggests that, on average, reducing SNF by 37% does not lead to a significant decrease

in yields, indicating that a 30% reduction is entirely feasible. This finding supports the feasibility of Japan's "Midori Target" of reducing chemical fertilizer use by 30%, which can be achieved with existing nitrogen management strategies. However, further testing and refinement are essential for each crop type and region due to the diversity of farmland practices across Japan. Nevertheless, we believe that the results of our meta-analysis provide a solid scientific basis for setting national goals, although there may be limited useful alternatives among other types of agricultural land management practices. It is crucial to conduct thorough initial surveys in country to determine farmers' actual SNF applications by cultivar and region and analyze the results to set realistic national targets.

Our study also demonstrated that achieving the Midori target alone will not significantly reduce waste nitrogen emissions from agricultural land. The benefits of reducing chemical fertilizers under the Midori Target are limited on a national scale when compared to the more ambitious goal of halving nutrient runoff to the hydrosphere, as outlined in Target 7 of the CBD (*The Convention on Biological Diversity, 2022*), using 2015 as the reference year. To achieve the reduction target, it will be necessary to increase NUE and implement reduction efforts in cooperation with non-agricultural sectors through integrated nitrogen management.

CRedit authorship contribution statement

Kazuya Nishina: Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ken-taro Hayashi:** Writing – review & editing, Supervision, Project administration. **Azusa Oita:** Writing – review & editing, Supervision. **Kei Asada:** Writing – review & editing, Supervision. **Atsushi Hayakawa:** Writing – review & editing, Supervision, Conceptualization. **Tomohiro Okadera:** Writing – review & editing. **Takashi Onodera:** Writing – review & editing. **Tatsuya Hanaoka:** Writing – review & editing, Funding acquisition. **Kazuaki Tsuchiya:** Writing – review & editing. **Koya Kobayashi:** Writing – review & editing, Software, Data curation. **Nobuhisa Koga:** Writing – review & editing, Software.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Kazuya Nishina reports financial support was provided by The Environment Research and Technology Development Fund. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study conducted as part of the JpNwst project, which was funded by the Environment Research and Technology Development Fund (JPMEERF20235001) of the Environmental Restoration and Conservation Agency provided by Ministry of the Environment of Japan. KN, KH, AO, and TO were also partially supported by the Research Institute for Humanity and Nature (RIHN: a constituent member of the NIHU), Japan Project No. RIHN14200156.

Data availability

The dataset (and reference list) used in meta-analysis in this study are available at figshare (DOI: [10.6084/m9.figshare.26808307](https://doi.org/10.6084/m9.figshare.26808307) and URL: <https://figshare.com/s/97b9008d63eb5839cd57>).

References

- Adams, D.C., Gurevitch, J., Rosenberg, M.S., 1997. Resampling tests for meta-analysis of ecological data. *Ecology* 78, 1277–1283.
- Akiyama, H., Sano, T., Nishina, K., Sudo, S., Oura, N., Fujimori, M., Uezono, I., Yano, S., Ohkoshi, S., Fujita, Y., et al., 2023. N₂O emission factors for organic amendments in Japan from measurement campaign and systematic review. *Sci. Total Environ.* 864, 161088. <http://dx.doi.org/10.1016/j.scitotenv.2022.161088>.
- Association of Agriculture and Forestry Statistics, 2023. *Pocket Fertilizer Handbook*. Norin Toukei Kyokai, Tokyo (in Japanese).
- Beig, B., Niazi, M.B.K., Jahan, Z., Hussain, A., Zia, M.H., Mehran, M.T., 2020. Coating materials for slow release of nitrogen from urea fertilizer: A review. *J. Plant Nutr.* 43, 1510–1533. <http://dx.doi.org/10.1080/01904167.2020.1744647>.
- Billen, G., Aguilera, E., Einarsson, R., Garnier, J., Gingrich, S., Grizzetti, B., Las-saletta, L., Le Noë, J., Sanz-Cobena, A., 2024. Beyond the farm to fork strategy: Methodology for designing a European agro-ecological future. *Sci. Total Environ.* 908, 168160. <http://dx.doi.org/10.1016/j.scitotenv.2023.168160>.
- Brownlie, W., Aziz, T., Sutton, M., Shen, K., Winkler, M., Robinson, G., Adhya, T., Amon, B., Awan, M., Bittman, S., Bleeker, A., Borghardt, G., Butterbach-Bahl, K., Brandani, S., Constantinescu, M., Dahl, K., Dalgaard, T., Delon, C., Döhler, H., Galy-Lacaux, C., Garnier, J., Groenestein, K., Hailemariam, M., Hayashi, K., Hillier, J., Jensen, L., Kawamoto, T., Kozlova, N., Lukin, S., Lund, T., Marques-dos Santos, C., Maqsood, M., Mathenge, C., Medinets, S., Minamisawa, K., Misselbrook, T., Nishina, K., Oenema, O., Pacheco, F., Parajuli, D., Rahman, M., Rees, R., Sanz-Cobena, A., Shiraz, H., Takahashi, A., Tanaka, H., Uchida, Y., Wakeel, A., Winiwarter, W., Yan, X., 2024. Nitrogen mitigation. In: INMS Guidance Document on Measures for Sustainable Nitrogen Management. International Nitrogen Management System, UK Centre for Ecology & Hydrology, Edinburgh, UK. <http://dx.doi.org/10.5281/zenodo.13831114>.
- Chen, Y., Camps-Arbestain, M., Shen, Q., Singh, B., Cayuela, M.L., 2018. The long-term role of organic amendments in building soil nutrient fertility: a meta-analysis and review. *Nutr. Cycl. Agroecosyst.* 111, 103–125. <http://dx.doi.org/10.1007/s10705-017-9903-5>.
- Dhakal, C., Lange, K., 2021. Crop yield response functions in nutrient application: A review. *Agron. J.* 113, 5222–5234. <http://dx.doi.org/10.1002/agg2.20863>.
- Elder, A., Fong, Y., 2019. Estimation and inference for upper hinge regression models. *Environ. Ecol. Stat.* 26, 287–302. <http://dx.doi.org/10.1007/s10651-019-00428-1>.
- Fong, Y., Huang, Y., Gilbert, P.B., Permar, S.R., 2017. chngpt: Threshold regression model estimation and inference. *BMC Bioinformatics* 18, 1–7. <http://dx.doi.org/10.1186/s12859-017-1863-x>.
- Fuhrmann-Aoyagi, M.B., Miura, K., Watanabe, K., 2024. Sustainability in Japan's agriculture: An analysis of current approaches. *Sustainability* 16, <http://dx.doi.org/10.3390/su16020596>.
- Gu, B., Leach, A.M., Ma, L., Galloway, J.N., Chang, S.X., Ge, Y., Chang, J., 2013. Nitrogen footprint in China: food, energy, and nonfood goods. *Environ. Sci. Technol.* 47, 9217–9224. <http://dx.doi.org/10.1021/es401344h>.
- Gu, B., Zhang, X., Lam, S.K., Yu, Y., van Grinsven, H.J.M., Zhang, S., Wang, X., Bodirsky, B.L., Wang, S., Duan, J., Ren, C., Bouwman, L., de Vries, W., Xu, J., Sutton, M.A., Chen, D., 2023. Cost-effective mitigation of nitrogen pollution from global croplands. *Nature* 613, 77–84. <http://dx.doi.org/10.1038/s41586-022-05481-8>.
- Hayashi, K., Shibata, H., Oita, A., Nishina, K., Ito, A., Katagiri, K., Shindo, J., Winiwarter, W., 2021. Nitrogen budgets in Japan from 2000 to 2015: Decreasing trend of nitrogen loss to the environment and the challenge to further reduce nitrogen waste. *Environ. Pollut.* 286, 117559.
- Hayashi, K., Tokida, T., Arai, M., Sakai, H., Nakamura, H., Hasegawa, T., 2022. Fertilizer-derived nitrogen use of two varieties of single-crop paddy rice: a free-air carbon dioxide enrichment study using polymer-coated ¹⁵N-labeled urea. *Soil Sci. Plant Nutr.* 68, 41–52. <http://dx.doi.org/10.1080/00380768.2021.2003163>.
- Hijbeek, R., van Ittersum, M.K., ten Berge, H.F., Gort, G., Spiegel, H., Whitmore, A.P., 2017. Do organic inputs matter—a meta-analysis of additional yield effects for arable crops in Europe. *Plant Soil* 411, 293–303. <http://dx.doi.org/10.1007/s11104-016-3031-x>.
- Imada, Y., Kawase, H., Watanabe, M., Arai, M., Shioyama, H., Takayabu, I., 2020. Advanced risk-based event attribution for heavy regional rainfall events. *NPG Clim. Atmos. Sci.* 3, 37. <http://dx.doi.org/10.1038/s41612-020-00141-y>.
- JA, 2024. Japan Agricultural cooperatives (JA) homepage. URL <https://www.zennoh.or.jp/>.
- Jiang, S., Hua, H., Jarvie, H.P., Liu, X., Zhang, Y., Sheng, H., Liu, X., Zhang, L., Yuan, Z., 2018. Enhanced nitrogen and phosphorus flows in a mixed land use basin: drivers and consequences. *J. Clean. Prod.* 181, 416–425. <http://dx.doi.org/10.1016/j.jclepro.2018.02.005>.
- Katsumi, N., Kusube, T., Nagao, S., Okochi, H., 2020. The role of coated fertilizer used in paddy fields as a source of microplastics in the marine environment. *Mar. Pollut. Bull.* 161, 111727. <http://dx.doi.org/10.1016/j.marpolbul.2020.111727>.
- Koga, N., Kobayashi, S., Mochizuki, K., 2023a. Application for computing nitrogen mineralization from organic amendments in soils. *Jpn. J. Soil Sci. Plant Nutr.* 94, 454–456. http://dx.doi.org/10.20710/dojo.94.6_454 (in Japanese).

- Koga, N., Nishina, K., Mochizuki, K., Kobayashi, S., Niimi, H., Ihara, H., Yamaguchi, N., Yamane, T., Kusaba, T., 2023b. A statistical model with acid detergent-soluble organic nitrogen content of organic amendments predicts nitrogen mineralization of various organic amendments in soil. *Jpn. J. Soil Sci. Plant Nutr.* 94, 106–114. <http://dx.doi.org/10.20710/dojo.94.2.106> (in Japanese).
- Lassaletta, L., Billen, G., Grizzetti, B., Anglade, J., Garnier, J., 2014. 50 year trends in nitrogen use efficiency of world cropping systems: the relationship between yield and nitrogen input to cropland. *Environ. Res. Lett.* 9, 105011. <http://dx.doi.org/10.1088/1748-9326/9/10/105011>.
- Lassaletta, L., Einarsson, R., Quemada, M., 2023. Nitrogen use efficiency of tomorrow. *Nat. Food* 4, 281–282. <http://dx.doi.org/10.1038/s43016-023-00740-x>.
- Leip, A., Caldeira, C., Corrado, S., Hutchings, N.J., Lesschen, J.P., Schaap, M., de Vries, W., Westhoek, H., van Grinsven, H.J., 2022. Halving nitrogen waste in the European union food systems requires both dietary shifts and farm level actions. *Glob. Food Secur.* 35, 100648. <http://dx.doi.org/10.1016/j.gfs.2022.100648>, URL <https://www.sciencedirect.com/science/article/pii/S2211912422000384>.
- Leon, A., Kohyama, K., Mishima, S., Ohkura, T., Shirato, Y., Takata, Y., Taniyama, I., Obara, H., 2012. Factors controlling organic amendment application rate and long-term change in application rate in Japanese paddy field using longitudinal questionnaire survey dataset (the basic soil environment monitoring project, stationary monitoring, 1979–1998). *Soil Sci. Plant Nutr.* 58, 104–120. <http://dx.doi.org/10.1080/00380768.2012.655675>.
- Liu, S., Pubu, C., Zhu, Y., Hao, W., Zhang, G., Han, J., 2023. Optimizing nitrogen application depth can improve crop yield and nitrogen uptake – a global meta-analysis. *Field Crops Res.* 295, 108895. <http://dx.doi.org/10.1016/j.fcr.2023.108895>.
- Mishima, S., Kimura, S.D., Eguchi, S., Shirato, Y., 2012. Estimation of the amounts of livestock manure, rice straw, and rice straw compost applied to crops in Japan: a bottom-up analysis based on national survey data and comparison with the results from a top-down approach. *Soil Sci. Plant Nutr.* 58, 83–90. <http://dx.doi.org/10.1080/00380768.2012.658581>.
- Mishima, S., Kohyama, K., 2010. The database and the methodologies to estimate recent trend of nitrogen (N) and phosphate (P) flows and residual n and p in Japanese national prefectural scales and examples their application. *Bull. Natl. Inst. Agro-Environ. Sci.* 27, 117–139 (in Japanese with English Abstract).
- Mori, N., Takemi, T., Tachikawa, Y., Tatano, H., Shimura, T., Tanaka, T., Fujimi, T., Osakada, Y., Webb, A., Nakakita, E., 2021. Recent nationwide climate change impact assessments of natural hazards in Japan and East Asia. *Weather Clim. Extrem.* 32, 100309. <http://dx.doi.org/10.1016/j.wace.2021.100309>.
- Nagano prefecture, 2023. Nagano prefecture environmentally friendly agricultural technology collection (soil fertilizer) (in Japanese) URL <https://www.pref.nagano.lg.jp/nogi/gijutsushu-dojouhiryuu.html>.
- Nishina, K., Akiyama, H., Nishimura, S., Sudo, S., Yagi, K., 2012. Evaluation of uncertainties in N₂O and NO fluxes from agricultural soil using a hierarchical Bayesian model. *J. Geophys. Res. Biogeosci.* 117, G04008. <http://dx.doi.org/10.1029/2012JG002157>.
- OECD, 2023. Agricultural Policy Monitoring and Evaluation 2023. OECD publishing, Paris. <http://dx.doi.org/10.1787/b14de474-en>.
- Quemada, M., Lassaletta, L., 2024. Fertilizer dependency: a new indicator for assessing the sustainability of agrosystems beyond nitrogen use efficiency. *Agron. Sust. Dev.* 44, 44. <http://dx.doi.org/10.1007/s13593-024-00978-0>.
- Ren, C., Liu, S., Van Grinsven, H., Reis, S., Jin, S., Liu, H., Gu, B., 2019. The impact of farm size on agricultural sustainability. *J. Clean. Prod.* 220, 357–367.
- Richardson, K., Steffen, W., Lucht, W., Bendtsen, J., Cornell, S.E., Donges, J.F., Drüke, M., Fetzer, I., Bala, G., von Bloh, W., Feulner, G., Fiedler, S., Gerten, D., Gleeson, T., Hofmann, M., Huiskamp, W., Kummer, M., Mohan, C., Nogués-Bravo, D., Petri, S., Porkka, M., Rahmstorf, S., Schaphoff, S., Thonicke, K., Tobian, A., Virkki, V., Wang-Erlandsson, L., Weber, L., Rockström, J., 2023. Earth beyond six of nine planetary boundaries. *Sci. Adv.* 9, eadh2458. <http://dx.doi.org/10.1126/sciadv.adh2458>.
- Sanz-Cobena, A., Lassaletta, L., Rodríguez, A., Aguilera, E., Piñero, P., Moro, M., Garnier, J., Billen, G., Einarsson, R., Bai, Z., Ma, L., Puigdueta, I., Ruiz-Ramos, M., Vallejo, A., Zaman, M., Infante-Amate, J., Gimeno, B.S., 2023. Fertilization strategies for abating n pollution at the scale of a highly vulnerable and diverse semi-arid agricultural region (Murcia, Spain). *Environ. Res. Lett.* 18, 064030. <http://dx.doi.org/10.1088/1748-9326/acd709>.
- Shindo, J., Okamoto, K., Kawashima, H., Konohira, E., 2009. Nitrogen flow associated with food production and consumption and its effect on water quality in Japan from 1961 to 2005. *Soil Sci. Plant Nutr.* 55, 532–545. <http://dx.doi.org/10.1111/j.1747-0765.2009.00382.x>.
- Shoji, S., Delgado, J., Mosier, A., Miura, Y., 2001. Use of controlled release fertilizers and nitrification inhibitors to increase nitrogen use efficiency and to conserve air and water quality. *Commun. Soil Sci. Plant Anal.* 32, 1051–1070. <http://dx.doi.org/10.1081/CSS-100104103>.
- Snapp, S., Sapkota, T.B., Chamberlin, J., Cox, C.M., Gameda, S., Jat, M.L., Marenja, P., Mottaleb, K.A., Negra, C., Senthilkumar, K., Sida, T.S., Singh, U., Stewart, Z.P., Tesfaye, K., Govaerts, B., 2023. Spatially differentiated nitrogen supply is key in a global food–fertilizer price crisis. *Nat. Sustain.* 6, 1268–1278. <http://dx.doi.org/10.1038/s41893-023-01166-w>.
- Sutton, M.A., Bleeker, A., Howard, C., Erisman, J., Abrol, Y., Bekunda, M., Datta, A., Davidson, E., De Vries, W., Oenema, O., et al., 2013. Our Nutrient World. the Challenge To Produce more Food & Energy with Less Pollution. Technical Report, Centre for Ecology & Hydrology, URL <https://wedocs.unep.org/20.500.11822/10747>.
- Sutton, M., Drewer, J., Moring, A., Adhya, T., Ahmed, A., Bhatia, A., Brownlie, W., Dragosits, U., Ghude, S., Hillier, J., Hooda, S., Howard, C., Jain, N., Kumar, D., Kumar, R., Nayak, D., Neeraja, C., Prasanna, R., Price, A., Ramakrishnan, B., Reay, D., Singh, R., Skiba, U., Smith, J., Sohi, S., Subrahmanyam, D., Surekha, K., van Grinsven, H., Vieno, M., Voletti, S., Pathak, H., Raghuram, N., 2017. 2 - The Indian nitrogen challenge in a global perspective. In: Abrol, Y.P., Adhya, T.K., Aneja, V.P., Raghuram, N., Pathak, H., Kulshrestha, U., Sharma, C., Singh, B. (Eds.), *The Indian Nitrogen Assessment*. Elsevier, pp. 9–28. <http://dx.doi.org/10.1016/B978-0-12-811836-8.00002-1>.
- Sutton, M.A., Howard, C.M., Kanter, D.R., Lassaletta, L., Moring, A., Raghuram, N., Read, N., 2021. The nitrogen decade: mobilizing global action on nitrogen to 2030 and beyond. *One Earth* 4, 10–14. <http://dx.doi.org/10.1016/j.oneear.2020.12.016>.
- Tang, S., Cheng, W., Hu, R., Guigue, J., Hattori, S., Tawarayama, K., Tokida, T., Fukuoka, M., Yoshimoto, M., Sakai, H., Usui, Y., Xu, X., Hasegawa, T., 2021. Five-year soil warming changes soil c and n dynamics in a single rice paddy field in Japan. *Sci. Total Environ.* 756, 143845. <http://dx.doi.org/10.1016/j.scitotenv.2020.143845>.
- The Convention on Biological Diversity, 2022. Kunming–Montréal global biodiversity framework. URL <https://www.cbd.int/article/cop15-final-text-kunming-montreal-gbf-221222>.
- Tian, H., Bian, Z., Shi, H., Qin, X., Pan, N., Lu, C., Pan, S., Tubiello, F.N., Chang, J., Conchedda, G., Liu, J., Mueller, N., Nishina, K., Xu, R., Yang, J., You, L., Zhang, B., 2022a. History of anthropogenic nitrogen inputs (hani) to the terrestrial biosphere: a 5 arcmin resolution annual dataset from 1860 to 2019. *Earth Syst. Sci. Data* 14, 4551–4568. <http://dx.doi.org/10.5194/essd-14-4551-2022>.
- Tian, H., Bian, Z., Shi, H., Qin, X., Pan, N., Lu, C., Pan, S., Tubiello, F.N., Chang, J., Conchedda, G., Liu, J., Mueller, N., Nishina, K., Xu, R., Yang, J., You, L., Zhang, B., 2022b. History of anthropogenic nitrogen inputs (hani) to the terrestrial biosphere: a 5 arcmin resolution annual dataset from 1860 to 2019. *Earth Syst. Sci. Data* 14, 4551–4568. <http://dx.doi.org/10.5194/essd-14-4551-2022>, URL <https://essd.copernicus.org/articles/14/4551/2022/>.
- Tuomisto, H.L., Hodge, I.D., Riordan, P., Macdonald, D.W., 2012. Does organic farming reduce environmental impacts?—A meta-analysis of European research. *J. Environ. Manag.* 112, 309–320. <http://dx.doi.org/10.1016/j.jenvman.2012.08.018>.
- Valkama, E., Salo, T., Esala, M., Turtola, E., 2013. Nitrogen balances and yields of spring cereals as affected by nitrogen fertilization in Northern conditions: A meta-analysis. *Agric. Ecosyst. Environ.* 164, 1–13. <http://dx.doi.org/10.1016/j.agee.2012.09.010>.
- Wu, H., Yang, T., Liu, X., Li, H., Gao, L., Yang, J., Li, X., Zhang, L., Jiang, S., 2020. Towards an integrated nutrient management in crop species to improve nitrogen and phosphorus use efficiencies of Chaohu watershed. *J. Clean. Prod.* 272, 122765. <http://dx.doi.org/10.1016/j.jclepro.2020.122765>.
- Xue, C., Ye, C., Liu, P., Zhang, C., Su, H., Bao, F., Cheng, Y., Catoire, V., Ma, Z., Zhao, X., Lu, K., Liu, Y., McGillen, M., Mellouki, A., Mu, Y., 2022. Strong HONO emissions from fertilized soil in the North China plain driven by nitrification and water evaporation (preprint (version 3)). *Res. Sq.* <http://dx.doi.org/10.21203/rs.3.rs-2045348/v3>.
- Yamaki, A., 2011. Nitrate leaching from vegetable field in Shimousa plateau and the effect of decrease measures. *J. Jpn. Assoc. Hydrol. Sci.* 41, 71–78. <http://dx.doi.org/10.4145/jahs.41.71> (in Japanese).
- Zhang, X., Zou, T., Lassaletta, L., Mueller, N.D., Tubiello, F.N., Lisk, M.D., Lu, C., Conant, R.T., Dorich, C.D., Gerber, J., et al., 2021. Quantification of global and national nitrogen budgets for crop production. *Nat. Food* 2, 529–540. <http://dx.doi.org/10.1038/s43016-021-00318-5>.